Aditya Ingle CSE-AIDS PANEL-D Batch-D1

MPJ - Lab - 03

* Problem Statement

Write a Java Prog to implement method overloading and method overriding in Java

1) write a class Shapes. find area of basic shapes using polymorphism

2) write a Java prog to create parent class Hillstations and 3 sub-classes by hill station names. Subclass extend superclass and override methods famousFood() & famousFor(). Call methods famousFood() and famousFor() by parent class i.e. Hillstation class. It should refer to base class object & base class method overrides superclass method base class method is invoked at runtime

* Objectives

→ To study method overloading and overriding in Java

* Theory

→ Method overloading

It means defining multiple methods with same name but different parameters. It is an eg of compile-time polymorphism

Advantages

- Improve Code readability
- Allow reuse of method name
- Reduce code duplication

→ Method Overridding

P.T.O.

It occurs when a subclass provides specific implementation of method already defined in its parent class. It is runtime polymorphism

Rules

- Same method name & parameters
- Must be in inheritance
- Cannot reduce access level
- Return type should be same or compatible

* CONCLUSION

Thus we have implemented method overloading & method overridding in Java

* FAQ

Q1) Can we overload main() method?

→ Yes, main() can be overloaded, but JVM calls only public static void main(String[] args)

Q2) Can overloaded methods be final?

→ Yes, but they cannot be overridden

Q3) Can overloaded methods be overridden?

→ Yes, if it is inherited in subclass

Q4) What is method overriding?

→

P.T.O.

→ Redefining parent class method in child class

Q5) What happens if augmented change in overriding?

→ It becomes overloading, not overriding

Q6) Can we change return type?

→ Yes, only if it is covariant (eg. subclass type).